

Food Waste Redistribution System for Restaurants and NGOs

AaharSetu

Team Name: Quantum Crew

Members :	G. Varshith Raju	2503A51L02
	M. Sai Samhitha	2503A51L52
	K. Srikar	2503A51L03
	D. Siddhartha	2503A51L04

1. Introduction

Food wastage is a major issue in urban areas, especially in cities like Hyderabad and Mumbai. Restaurants, hotels, and event halls generate a large amount of surplus food daily. At the same time, many underprivileged communities struggle to access basic meals.

Despite the presence of NGOs working to reduce hunger, there is no efficient system to connect food donors with organizations that can distribute food quickly.

2. Problem Statement

The current food redistribution process is inefficient due to:

- Lack of a centralized platform
 - Manual communication via phone calls and messaging apps
 - Delay in food pickup and delivery
 - Poor coordination between restaurants, NGOs, and volunteers
 - No tracking system for food donations
 - Food getting spoiled before reaching beneficiaries
-

3. Key Challenges Identified

1. **Communication Gap**
No real-time communication system between stakeholders
 2. **Time Sensitivity**
Food has limited freshness duration
 3. **Logistics Issues**
Difficulty in arranging timely transportation
 4. **Lack of Data Tracking**
No proper record of donations and distribution
 5. **Volunteer Coordination**
Hard to identify available volunteers nearby
-

4. Target Users / Stakeholders

- **Restaurants / Event Organizers**
(Food donors)
 - **NGOs**
(Food collectors and distributors)
 - **Volunteers**
(Delivery agents)
 - **Admin**
(System controller)
-

5. Proposed Solution: AaharSetu

AaharSetu is a smart food redistribution platform that acts as a bridge between food donors and NGOs, ensuring efficient and timely delivery of surplus food through a coordinated volunteer network.

6. Core Idea

The system will:

- Allow restaurants to upload surplus food details
- Notify nearby NGOs in real-time
- Assign volunteers for pickup and delivery
- Track the entire process
- Store data for analysis and reporting

7. Objectives

- Reduce food wastage
 - Improve coordination between stakeholders
 - Ensure timely delivery of food
 - Provide transparency and tracking
 - Maximize utilization of surplus food
-

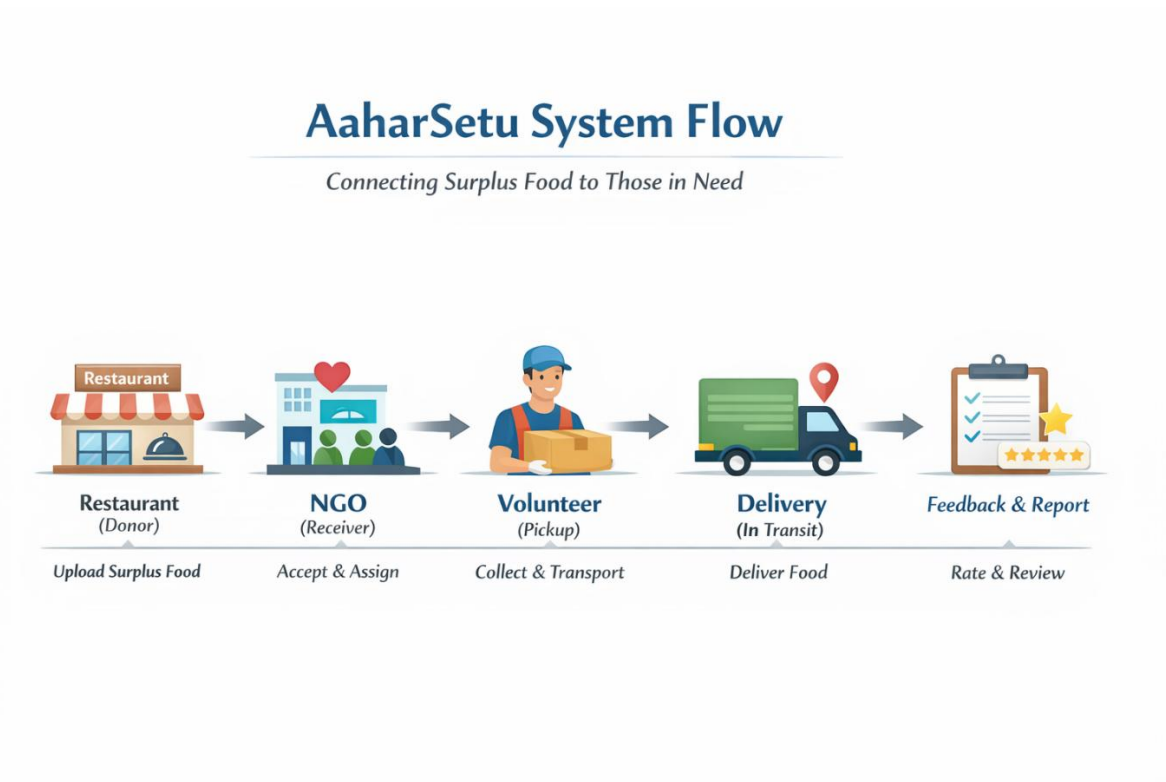
8. Brainstormed Solution Ideas

- Real-time notification system
 - Location-based matching of NGOs
 - Volunteer assignment system
 - Food freshness tracking
 - Emergency food request feature
 - Impact dashboard (meals saved, waste reduced)
-

9. System Flow (High-Level)

1. Restaurant uploads surplus food
2. System identifies nearby NGOs
3. NGO accepts the request
4. Volunteer is assigned
5. Food is picked up and delivered

6. Status is updated and stored
7. Feedback is collected



11. GitHub Repository Plan

Repository Name: **AaharSetu**

Structure:

- /frontend
- /backend
- /docs
- /presentation

Includes:

- Source code
- Documentation files
- Architecture diagram
- README file

12. Expected Outcome

A functional prototype of AaharSetu that:

- Demonstrates food donation flow
 - Shows interaction between all user roles
 - Tracks and displays delivery status
 - Highlights reduction in food wastage
-

13. Conclusion

AaharSetu aims to solve the critical issue of food wastage by creating a structured, efficient, and scalable system that connects surplus food sources with those in need.

It transforms unorganized efforts into a coordinated, technology-driven solution with real-world impact.
